🏗️ Your Complete Zero-to-Finance Learning Roadmap

Built for a Morgan Stanley Software Engineer Starting from Absolute Zero

> ***Your Situation:*** *Java full-stack job hunt + McDaniel M.S. Data Analytics (AI track) starting soon.*
> ***Your Advantage:*** *You built the systems at Morgan Stanley. Now you'll understand what they actually do.*
> ***Your Constraint:*** *~4–5 hours/week max. This roadmap respects that.*

How This Roadmap Works

Think of finance knowledge as a building with **four floors**. Most "beginner" courses (including Shiller's famous Yale course) assume you already live on Floor 1. You don't. Skipping floors costs you *more* time, not less — you'll spend half your study hours Googling vocabulary instead of learning concepts.

Floor	What You Learn	Time	Cost	When
Floor 0	How money actually works	~3–4 weeks	$0	Start immediately
Floor 1	Personal finance + basic accounting	~6–8 weeks	$0	After Floor 0
Floor 2	Markets, investing, and how the system fits together	~8–10 weeks	$0 (audit)	After Floor 1
Floor 3	Fintech, quant finance, and the AI crossover	~10–12 weeks	Coursera Plus	After Floor 2 + McDaniel coursework

Total timeline: ~6 months at 4–5 hours/week.

FLOOR 0: How Money Works (Weeks 1–4)

Goal: Stop feeling lost when you hear words like "inflation," "liquidity," "compound interest," or "credit score."

Primary Course

Khan Academy — Financial Literacy (free, no signup required)
khanacademy.org/college-careers-more/financial-literacy

This is a 16-unit, self-paced course. Do **every unit in order** — do not skip around. Each unit is 30–60 minutes of video + exercises.

Unit	Topic	Why It Matters for You
1	Welcome to Financial Literacy	Mindset framing
2	Budgeting and Saving	50/30/20 rule, expense tracking
3	Consumer Credit	**Credit scores, credit cards, payment methods** — start here if impatient
4	Financial Goals	SMART goals, net worth calculation
5	Loans and Debt	Borrowing terms, bankruptcy basics
6	Insurance	Health, life, property, disability
7	Investments and Retirement	**Introduction to risk/return, saving vs. investing**
8	Scams & Fraud	Protect yourself
9	Careers and Education	Earnings potential, education ROI
10	**Taxes and Tax Forms**	**W-2, 1099, 1040, brackets, deductions** — critical for your immigration/tax situation
11	Employment	Paycheck anatomy, benefits, compensation structures
12	**Banking**	**How banks create money, interest, inflation** — connects directly to Morgan Stanley's business
13	Car Buying	Loans, leasing vs. owning
14	Housing	Renting vs. buying, mortgages, lease agreements
15	Teacher Resources	Skip
16	Additional Resources	Supplementary reading

Time commitment: 2–3 hours/week for 3–4 weeks.

Pro tip: After the Banking unit (Unit 12), spend 20 minutes reading about how the Federal Reserve works. You processed Fedwire transactions at Morgan Stanley — now you'll know what they *are*.

FLOOR 1: Personal Finance + Basic Accounting (Weeks 5–12)

Goal: Read a company's financial statements and understand your own tax situation. Build the vocabulary needed for everything that comes next.

Course 1A: Personal Finance & Credit (Free)

Alison — An Introduction to Personal Finance and Credit
alison.com/course/an-introductory-course-in-personal-finance-and-credit

- **Duration:** 6–10 hours
- **Cost:** Free (optional paid certificate)
- **What you'll learn:** Budgeting, asset vs. liability, good vs. bad debt, loan types, compound interest, accessing credit
- **Why this one:** Short, practical, and fills gaps Khan Academy leaves. The compound interest module is especially strong.

Course 1B: Financial Accounting (Free to Audit)

Wharton — Introduction to Financial Accounting (Coursera)
coursera.org/learn/wharton-accounting

- **Duration:** ~4 weeks at 4–6 hours/week
- **Cost: Free to audit** (you have Coursera paid, but auditing is fine here)
- **What you'll learn:**
- The balance sheet equation: Assets = Liabilities + Stockholders' Equity
- Debit/credit bookkeeping and T-accounts
- How to read the three core statements: Balance Sheet, Income Statement, Cash Flow Statement
- Accrual accounting vs. cash accounting
- Revenue recognition and expense matching
- **Why this one:** This is the course that will retroactively explain what Morgan Stanley was *actually doing* with the financial systems you built. When you see "general ledger" or "reconciliation" in a job description now, you'll know what it means.
- **Prerequisite:** Floor 0 complete (you need to know what a bank does and what interest is).

Course 1C: U.S. Federal Taxation (Coursera Paid — Use Your Subscription)

University of Illinois — Federal Taxation I: Individuals, Employees, and Sole Proprietors
coursera.org/learn/federal-taxation-individuals

- **Duration:** ~3 weeks at 10 hours/week (or 6 weeks at 5 hours/week)
- **Cost:** Included in your Coursera Plus subscription
- **Rating:** 4.8/5 stars, 97% of learners recommend it
- **What you'll learn:**
- Gross income, inclusions, and exclusions
- Deductions (for-AGI and from-AGI)
- Employee vs. self-employed tax treatment
- Retirement plans (Traditional IRA, Roth IRA, 401k)
- **Hands-on practice completing an actual Form 1040**
- **Why this one specifically for you:** With your immigration situation (F-1 CPT, H-1B history, potential mixed employment types), understanding employee vs. self-employed tax distinctions is *directly useful*, not just theoretical. This course was recently updated for current tax legislation.
- **Approach:** Do 30 minutes of Khan Academy's Taxes unit (Unit 10) first to learn the vocabulary, then take this course.

Floor 1 Total Time: ~6–8 weeks at 4–5 hours/week.

FLOOR 2: Markets & Investing Basics (Weeks 13–22)

Goal: Understand what stocks and bonds actually are, how markets work, what risk means quantitatively, and how portfolios are built. This is where you finally touch the "fintech" layer.

Course 2A: Financial Markets (Free to Audit)

Yale University — Financial Markets by Robert Shiller (Nobel Laureate)
coursera.org/learn/financial-markets

Also available at: online.yale.edu/courses/financial-markets

- **Duration:** 7 weeks, ~33 hours total
- **Cost: Free to audit** (certificate is $49, or included in Coursera Plus)
- **What you'll learn:**
- Behavioral finance and market bubbles
- Banking, debt, and equity
- Risk management and regulation
- Forward/futures markets and options
- Monetary policy and the Fed
- Innovation in finance (fintech, blockchain, AI in finance)
- **Why this one:** Widely considered one of the best finance courses ever made. Shiller is a Nobel Prize winner and genuinely engaging. But — and this is critical — **do NOT start this course before completing Floors 0 and 1**. Shiller uses terms like "yield," "leverage," "liquidity," and "equity" without defining them from zero. You'll drown without the foundation.
- **Your unique edge:** When Shiller talks about trade processing, settlement, or clearing, you'll have *concrete mental images* from Morgan Stanley. Most learners are guessing. You know what the systems look like.

Course 2B: Stocks and Bonds (Free)

Stanford Graduate School of Business — Stocks and Bonds: Risks and Returns
Stanford GSB Online

- **Duration:** ~5 sections, ~45 min video + 1–2 hours exercises per section
- **Cost:** Free, self-paced
- **What you'll learn:**
- How stocks and bonds are valued
- Interest rates and bond valuation
- Corporate and municipal bonds
- Tradeoffs in portfolio construction
- Includes a panel discussion with Nobel laureate William Sharpe
- **Why this one:** Stanford-quality, free, and focuses exactly on the gap between "I know what a stock is" and "I understand how to think about a portfolio." Perfect complement to Shiller's broader view.

Course 2C: Investing in Stocks (Udemy — Free via Library Card)

Udemy — Investing In Stocks The Complete Course! (17+ Hours) by Steve Ballinger
Available through your library's Udemy access

- **Duration:** 17 hours (do selectively after Stanford course)
- **Cost:** Free via your library card
- **What you'll learn:**
- Stock screeners and fundamental analysis
- Dividend stocks, growth stocks, value stocks
- Order types (market, limit, stop-loss)
- Risk metrics (Standard Deviation, Beta)
- ETFs and mutual funds
- **Why this one:** Practical, hands-on, and fills the "how do I actually buy a stock?" gap that academic courses leave.

Use it as a reference after Stanford, not a primary course.

Floor 2 Total Time: ~8–10 weeks at 4–5 hours/week.

FLOOR 3: Fintech, Quantitative Finance & The AI Crossover (Weeks 23–34)

Goal: This is where your McDaniel M.S. in Data Analytics (AI track) and your finance knowledge start reinforcing each other instead of competing. Quantitative finance is essentially machine learning applied to time-series financial data.

Course 3A: Economics of Money and Banking (Free to Audit)

Columbia University — Economics of Money and Banking by Perry Mehrling
Coursera

- **Duration:** ~13 weeks, but you can move faster selectively
- **Cost:** Free to audit
- **Rating:** 4.8/5
- **What you'll learn:**
- The hierarchy of money
- Shadow banking and dealer banks
- Monetary policy implementation
- The money view vs. the finance view
- How the plumbing of the financial system actually works
- **Why this one:** Mehrling teaches you to *think* like the financial system. This is the course that connects your software engineering background to high finance. When he discusses payment systems, clearing, and settlement, you'll understand the infrastructure because you *built* similar infrastructure.

Course 3B: Investment Management with Python and Machine Learning (Coursera Paid)

EDHEC Business School — Investment Management with Python and Machine Learning Specialization
coursera.org/specializations/investment-management-python-machine-learning

- **Duration:** ~4 months at 5 hours/week
- **Cost:** Included in Coursera Plus
- **What you'll learn:**
- Portfolio optimization with Python
- Risk management and factor models
- Machine learning for asset pricing
- Algorithmic trading strategies
- Backtesting and performance evaluation
- **Why this one:** This is the bridge. You're learning Python for data analytics at McDaniel. This specialization shows you how to aim that skill at finance. By this point, your finance vocabulary is solid and your Python is growing — this is where they meet.

Course 3C: Machine Learning for Trading (Coursera Paid)

Google Cloud & NYIF — Machine Learning for Trading Specialization
coursera.org/specializations/machine-learning-trading

- **Duration:** ~3 months

- **Cost:** Included in Coursera Plus
- **Prerequisites:** Intermediate Python, basic ML, basic finance markets knowledge
- **What you'll learn:**
- Quantitative trading strategies
- Pairs trading, momentum trading
- Deep learning and reinforcement learning for trading
- Using Google Cloud for scalable model deployment
- **Why this one:** More advanced than EDHEC. Take this *after* EDHEC and after you've completed at least one ML course in your McDaniel program.

Floor 3 Total Time: ~10–12 weeks at 5–6 hours/week (can overlap with McDaniel coursework).

🏷️ Quick-Reference: Your Course List by Platform

Platform	Course	Floor	Cost
Khan Academy	Financial Literacy (16 units)	0	Free
Alison	An Introduction to Personal Finance and Credit	1	Free
Coursera (audit)	Wharton — Introduction to Financial Accounting	1	Free
Coursera (paid)	Illinois — Federal Taxation I	1	Included in Plus
Coursera (audit)	Yale — Financial Markets (Shiller)	2	Free
Stanford (web)	Stocks and Bonds: Risks and Returns	2	Free
Udemy (library)	Investing In Stocks Complete Course	2	Free (library)
Coursera (audit)	Columbia — Economics of Money and Banking (Mehrling)	3	Free
Coursera (paid)	EDHEC — Investment Management with Python & ML	3	Included in Plus
Coursera (paid)	Google/NYIF — Machine Learning for Trading	3	Included in Plus

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⏰ Suggested Weekly Schedule (While Job Hunting + McDaniel)

Phase	Weeks	Weekly Hours	What to Do
Phase 1: Foundation	1–4	2–3 hrs	Khan Academy Financial Literacy only. Light enough to do passively. Swap one podcast/YouTube session per week for this.
Phase 2: Core Skills	5–12	4–5 hrs	Alison personal finance + Wharton accounting (alternate weeks). Add Illinois Taxation course in weeks 9–12.
Phase 3: Markets	13–22	4–5 hrs	Yale Shiller (primary) + Stanford Stocks & Bonds (secondary). Reference Udemy investing course as needed.
Phase 4: Integration	23–34	5–6 hrs	Mehrling's Money & Banking + EDHEC Python/ML specialization. This overlaps with McDaniel AI coursework — they reinforce each other.
Phase 5: Advanced	35+	5–6 hrs	Google/NYIF Machine Learning for Trading. Only after McDaniel ML fundamentals are solid.

💰 Cost Summary

Item	Cost
Khan Academy	$0
Alison	$0
Wharton Accounting (audit)	$0
Stanford Stocks & Bonds	$0
Illinois Taxation	Included in Coursera Plus
Yale Financial Markets (audit)	$0
Udemy (library card)	$0
Columbia Money & Banking (audit)	$0
EDHEC Python/ML Specialization	Included in Coursera Plus
Google/NYIF ML for Trading	Included in Coursera Plus
Total out-of-pocket	**$0 beyond your existing Coursera Plus subscription**

Immediate Action Items (This Week)

1. **Today:** Go to Khan Academy → Financial Literacy → Unit 3 (Consumer Credit) and Unit 10 (Taxes). Watch the Credit Score and "What are taxes?" videos. Total time: 30 minutes. These two concepts explain 80% of financial mistakes people make.
2. **This week:** Create a free Khan Academy account and bookmark the Financial Literacy course. Commit to 2 hours this week (Units 1–3).
3. **This week:** Check your library's Udemy access and confirm you can log in. Bookmark the Investing in Stocks course for later.
4. **Before McDaniel starts:** Complete at least Floor 0 and Floor 1. You'll enter your master's program with financial literacy that most data scientists lack — a genuine competitive advantage.

Why This Sequence Matters (The "Don't Skip Floors" Lecture)

The fintech/quant roadmaps you see online assume you already know: - What a bank does - What interest is and how it compounds - What a stock represents - What debt vs. equity means - What a balance sheet is - What inflation does to money - What credit scores measure

None of that is in the advanced courses. It's all assumed background. If you hit Shiller's course in Week 1 without this foundation, you'll spend half your time googling terms instead of learning concepts. The 2 months you "save" by skipping ahead will cost you 3 months of confusion later.

Your Morgan Stanley experience is a superpower here — but only *after* you have the vocabulary to map theory to the systems you built.

📚 Supplementary Resources (Optional, Free)

- **Podcast:** *Planet Money* (NPR) — 20-minute episodes explaining current financial events. Perfect for commutes.
- **YouTube:** *The Plain Bagel* — unbiased, high-quality investing explainers.
- **Book:** *A Random Walk Down Wall Street* by Burton Malkiel — read during Floor 2.
- **Book:** *The Big Short* by Michael Lewis — read during Floor 3 for context on 2008 (Shiller and Mehrling both reference it).
- **News:** *The Wall Street Journal* or *Financial Times* — start reading headlines during Floor 2. Don't worry about understanding everything; just build pattern recognition.

Roadmap built for a Morgan Stanley software engineer with access to Coursera Plus, Pluralsight, Udemy (library), and free platforms. Timeline assumes 4–5 study hours per week alongside full-time job hunting and graduate coursework.